University of Bahrain

College of Information Technology
Department of Computer Science

ITCS347 Design and Analysis of Algorithms

Second Semester 2025/2026
Home Assignment #2

Student Name		GRADE
Student ID		
Section		

Civil Engineering – Resultant Force of Components
When multiple forces act at different angles, the resultant force is:

$$F = \sqrt{(\sum F_x)^2 + (\sum F_y)^2}$$

where F_x and F_y are horizontal and vertical components stored in two arrays $Fx[0..n-1]$ and $Fy[0..n-1]$ of real numbers respectively.

1. Design a divide-and-conquer algorithm $\textsc{SumArray}(A, left, right)$ that computes the sum of all elements in a subarray $A[left..right]$.
2. Using $\textsc{SumArray}$, design an algorithm $\textsc{ComputeResult}(Fx, Fy, n)$ that computes the resultant force F.
3. Analyze the algorithm and find the time complexity recurrence relation $T(n)$.
4. Prove that $T(n)$ is linear (for the algorithm $\textsc{SumArray}$ use the *Master Theorem*).